W3 · Classification and Generalization

Logistic regression · Decision boundaries · Overfitting · Regularization

Name _____ Date _____

1. Concept Check

Choose the single best answer for each question.

1. What kind of task is classifying email as spam or legitimate?

(1) Predicting a continuous house price
(2) Converting input units
(3) Binary classification
(4) Counting examples

2. What is the output range of the sigmoid function for a finite input?

(1) Less than -1
(2) Between 0 and 1
(3) Always greater than 1
(4) Exactly 0 for every input

3. The predicted positive-class probability is 0.8, and the threshold for a positive prediction is 0.5. What is the result?

(1) Negative
(2) The probability cannot be calculated
(3) The actual target must be positive
(4) The model predicts the positive class

4. You raise the positive-class threshold from 0.5 to 0.8 while keeping the same predicted probabilities. Which statement is correct?

(1) The number of positive predictions cannot increase
(2) All predicted probabilities automatically increase
(3) Accuracy must increase
(4) All actual targets change

5. What is a decision boundary?

(1) The boundary of the folder containing the data
(2) The time when training ends
(3) A boundary in input space where the predicted class changes
(4) A line that randomly divides the targets

1. Concept Check / Continued

Choose the single best answer for each question.

6. For an example whose actual target is 1, the model predicts a positive probability of 0.01. Its logistic loss is greater than which alternative?

(1) Predicting a probability of 0.001
(2) Predicting a probability of 0.9
(3) Predicting the same probability of 0.01
(4) The loss is identical regardless of the probability

7. A model fits its training data very well but often fails on unseen data. What should you suspect?

(1) Necessarily perfect generalization
(2) A file format that stores too little data
(3) Proof that classification is impossible
(4) Overfitting

8. How can regularization help reduce overfitting?

(1) By penalizing excessively large weights and discouraging overly complex fitting
(2) By deleting all training targets
(3) By rewarding predictions with large errors
(4) By making every model perfect

9. With other conditions fixed, what can happen if the regularization strength λ is too large?

(1) Every error must disappear
(2) The number of input features automatically increases
(3) Weights can be suppressed so strongly that useful patterns are missed
(4) The number of actual targets increases

10. What is the best way to judge whether a classifier has learned successfully?

(1) Look only at training accuracy
(2) Also evaluate performance on data not used for training
(3) Always prefer larger weights
(4) Change every prediction to the same class

2. Think Like an ML Scientist

Question 11 · Read the pseudocode and find why the model does not work as intended.

Spam has y=1; legitimate email has y=0. The model receives a smaller loss when it is more confidently wrong. Assume the other calculations and learning rate are appropriate, every p lies strictly between 0 and 1, and log is the natural logarithm.

```
z = dot(x, w) + b
p = sigmoid(z)
loss = y * log(p) + (1 - y) * log(1 - p)
update_parameters_to_minimize(loss)
```

11(1) Identify what is missing from the loss definition and rewrite that line correctly.

11(2) For y=1, compare p=0.9 and p=0.1 to explain why the original code prefers incorrect predictions. Use log(0.9) ≈ -0.105 and log(0.1) ≈ -2.303.

3. Open Question

Question 12 · There is no single correct answer. Describe your method and reasoning.

You have a tool that converts sentences into lists of numbers and a small set of reviews labeled positive or negative. You want to classify unseen reviews and discover new patterns among reviews. In what order would you run your experiments?

Explain how you would use labeled and unlabeled reviews, check your results, and handle reviews the model is uncertain about. You may combine methods from the course or propose a new idea.

Equations and code are optional. You may answer with text, a diagram, or a simple flowchart.